Davis Liu

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Education

University of Waterloo, Bachelor of Software Engineering *Waterloo, ON*
94% Cumulative GPA *Sept 2024 – Present*

Experience

Programming Lead – Unionville High School VEX Robotics Team *Markham, ON Sept 2022 – June 2024*

- **Led software development** for operator-controlled and **fully autonomous** driving on **3 independent subteams**
- **Mentored junior members** in software development, physical design, and CAD
- Developed and maintained an **adaptable C++ programming library** for autonomous feedback control, on-screen selection menus, and debugging, which was used in **6 competition events over 2 years**
- Collaborated with mechanical and CAD divisions on designing and building

Projects

DevSearch *(Express, MySQL, TypeScript, Python, HTML/CSS)* *Jan 2024 – Current*

- Created a **search engine and web crawler** that scrapes and classifies software developer jobs from web pages
- Integrated MySQL database with Express backend to store scraped data

DoorSense *(React Native, TensorFlow, Python)* *Oct 2024 – Dec 2024*

- Collaboratively built a **smart door system** that supports doorstep monitoring, weather displays, and mobile notifications
- Leveraged **custom-trained TensorFlow models** to identify visitors, package deliveries, and potential intruders
- Integrated smart door system with a React Native mobile app to control smart door functionality

ShopSmart *(React Native, TypeScript)* *Oct 2024 – Current*

- Developed backend for a **React Native mobile app** that helps users optimize shopping trips based on prices and distances
- Applied **Dijkstra's algorithm with bitmasks** to compute an optimal way to purchase groceries

GuideBot — Hack the 6ix 2023 *(Arduino, C++, JavaScript, Tensorflow)* *Aug 2023*

- Utilized **embedded robotics software and object recognition** to create an urban-center navigation device for visually impaired individuals
- Successfully implemented pathfinding algorithms for indoor spaces by recognizing surrounding objects and walls
- Won award for **"Best Accessibility Hack"**

UACC Contest *(C++)* *Jan 2024*

- Co-authored a competitive programming contest hosted on the DMOJ judge, written by **170+ contestants**

Skills

Languages: C++, Java, JavaScript, TypeScript, HTML/CSS, C, Python, Lua
Technologies: Embedded Software, TensorFlow, Express, React, SDL2, Computer-Aided Design
Non-Technical: Leadership, Problem Solving, Mandarin Chinese

Awards

Euclid Mathematics Contest - Distinction	Score within the top 14% in 2023	2023, 2024
Canadian Computing Competition - Distinction	Score within the top 6.5% in 2023	2023, 2024
Canadian Open Mathematics Competition - Distinction		2023